

Econ 101A

Section 3

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1 Utility Functions

Definition: A utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$ if, for every pair of points x and y in X , $x \succsim y$ if and only if $u(x) \geq u(y)$. In other words, if x has higher utility than y , you prefer x to y .

- **Theorem (Existence):** If preference relation $\succsim$ is rational and continuous on set X , there exists a continuous utility function $u : X \rightarrow \mathbb{R}$ that represents it.
- **Non-uniqueness:** The utility function representing preferences is not unique. If $u(x)$ represents preferences $\succsim$ and f is a strictly increasing function, then $f(u(x))$ represents $\succsim$ as well. For most of the time we only care about the **ordinal** property of utility function.

2 Univariate Optimization

- Optimization is about finding optimums, i.e. maximums and minimums, of *scalar* functions.
- Recall that maximizing a function is equivalent to minimizing its negative. Thus:

$$\operatorname{argmax}_{x \in X} f(x) = \operatorname{argmin}_{x \in X} -f(x)$$

where $\operatorname{argmax}_{x \in X} f(x)$ ($\operatorname{argmin}_{x \in X} f(x)$) gives the solutions to the problem $\max_{x \in X} f(x)$ ($\min_{x \in X} f(x)$).

- A stationary point, or critical point, is a point at which a function is neither going up nor down - i.e. an x_0 at which $f'(x_0) = 0$. In practice finding an optimum usually means finding a stationary point, but note that being a stationary point is neither a necessary nor a sufficient condition for being an optimum, since we could have a corner solution.
- Considering only interior solutions (ignoring corner solutions), what must be true at a local maximum(minimum)?
 - FOC: The function is at a stationary point: $f'(x^*) = 0$
 - SOC: On either side of the stationary points, the function must be going down(up): $f''(x^*) < 0$ ($f''(x^*) > 0$).

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2.1 Necessary and Sufficient Conditions in the Context of Optimization

- FOC is a necessary condition for an **interior local** optimum.
- SOC is a necessary condition for an **interior local** optimum.
- Neither, taken alone, is a sufficient condition for an **interior local** optimum.
- FOC and SOC together are necessary and sufficient conditions for an **interior local** optimum.

2.2 Optimization of Concave Functions

Usually, to find global maximum (if it exists) of a function in a given range, we need to find all local maxima and pick the largest one. With concavity, it suffices to find the only stationary point with FOC.

- Definition: Function f from a convex set $C \subset R^n$ to R is concave (convex) if

$$f(tx + (1-t)y) \geq (\leq) tf(x) + (1-t)f(y)$$

for all $x, y \in C$ and for all $t \in [0, 1]$.

- A more useful characterization: suppose $f(x) : R \rightarrow R$ is twice differentiable, then f is concave iff $f'' \leq 0$ for **all** x .
- A useful result: Consider a **twice-differentiable concave** function over $C \subset R^n$. If x_0 satisfies FOC, then it is a global maximum.
 - Not applicable to cases where f'' is not globally identified (e.g. $f(x) = -|x|$, $x \in [-1, 1]$)
 - This is a **sufficient** condition for global maximum. Corner solutions are possible if no points satisfy FOC in the given range (e.g. $f(x) = -x^2$, $x \in [1, 2]$. What if $x \in [-1, 0) \cup (0, -1]$?)

3 Exercises

3.1 Problem 1 from PS1, Spring 2014

Consider the following maximization problem: $\max_x f(x; x_0) = \exp(-(x - x_0)^2)$

1. Write down the first order conditions for this problem with respect to x (notice that x_0 is a parameter).
2. Solve explicitly for the x^* that satisfies the first order conditions.
3. Now compute the second order conditions at x^* . Is the stationary point that you found in (2) a local maximum? Why or why not?
4. Is the function f concave in x ?

3.2 Problem 2 from PS1, Spring 2017

Consider the following maximization problem: $\max_x f(x; p) = (p - x)(x - 1)$. p is a parameter.

1. Write down FOC with respect to x . Solve explicitly for x^* that satisfies FOC.
2. Compute SOC. Is the stationary point that you found in part 2 a local maximum?
3. Is function f concave in x ? Is x^* a global maximum?
4. Comparative statics: compute dx^*/dp .
5. Now we want to verify Envelope Theorem (Cecile will prove this theorem in next class):
 - Plug x^* back in function f and compute $f(x^*(p); p)$ (so now there's no x in the expression of f). Then, take the derivative with respect to p .
 - Let's try a second way: first, treat f as a function of p and directly write down FOC w.r.t. p (you can treat x as a parameter here). Then, plug x^* in FOC. Compare your answer to the previous part.